Introduction to Software Design Thinking

Module 1 (09)

216U01E701
Software Design Thiniking
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Module 1: Introduction to Software Design Thinking

- Concept of design thinking in Software Engineering
- Stages: Empathize, Define, Ideate, Prototype, Test
- Human-centered software design approach
- Design vs. traditional problem-solving
- Case studies on successful design thinking in software

Design Thinking in Software Engineering

- **Designing a product:**

  Building a home (Stakeholders):
  - **Civil Engineer**
  - **Architect**
  - **Interior Designer**
  - **Home Owner**
  - **End users (House Wife, Children, Elder People)**

Design Thinking in Software Engineering

Stake Holder	POV (point of View) Interest
Civil Engineer	Stability, Safety
Architect	Standards
Interior Decorator	Aesthetics
Home Owner	Return on Investment
House Wife	Efficiency, maintenance, management
Children	Facilities, Entertainment
Elderly People	Accessibility, Comfort

Design Thinking in Software Engineering

- **Designing a product will engage:**

  An interdisciplinary team of designers, behavioural scientists, marketers, and engineers whose task was to identify appropriate constraints for the project.

- Involved in the project as it passed from inspiration through ideation and on into the implementation phase.

Design Thinking in Software Engineering

- **Three spaces of innovation of a Product**
  - **Inspiration:** the problem or opportunity that motivates the search for solutions
  - **Ideation:** the process of generating, developing, and testing ideas
  - **Implementation:** the path that leads from the project room to the market.

Performed in iterative manner

Design Thinking in Software Engineering

- **Three spaces of innovation of a Product**
  - **Inspiration:**
    - The *exploratory phase* where the design team immerses themselves in the world of the user to uncover problems, gather insights, and identify latent needs.
  - **Core Objectives of Inspiration**
    - **Divergent Exploration:** Moving away from immediate assumptions to discover new possibilities/ options.
    - **Uncovering Latent Needs:** Identifying requirements that users themselves cannot articulate because they are so accustomed to compensating for poor design.
    - **Shifting Perspective:** Moving from a *technocentric* focus ("What can our engineers build?") to a *human-centered* focus ("What do people actually need?").

Design Thinking in Software Engineering

- **Three spaces of innovation of a Product**
  - **Inspiration:**
  - **Approach:**
    - **Watch *What they DO:*** Will help in identifying how people use the product, where the end-user stuck …
    - **Listen to *what they SAY*:** understand the user's emotional state, frustrations, and motivations.
    - **Experience what they *FEEL*:** physically place themselves in the user's environment or temporarily take on their physical constraints
  - **Target the Extremes:** Will expose the pain points, flaws which will help in providing better solution

Design Thinking in Software Engineering

- **Three spaces of innovation of a Product**
  - **Ideation:** ***Quantity over Quality***
  - one problem many solutions
  - Don’t restrict to only one (few) solution but try to find as many as possible solutions

Design Thinking in Software Engineering

- **Three spaces of innovation of a Product**
  - **Implementation:** the path that leads from the project room to the market.

Lecture 2

- Stages: Empathize, Define, Ideate, Prototype, Test

The 5-Stage Framework

Empathize: Understand user needs and research their real-world problems.

Define: Synthesize research and frame a clear, actionable problem statement.

Ideate: Generate a wide range of creative solutions without initial judgment

Prototype: Build low-fidelity versions of ideas to quickly learn and iterate.

Test: Gather feedback from real users to validate and refine the concept.

The 5-Stage Framework



Empathize

Understand user needs and research their real-world problems.



Define

Synthesize research and frame a clear, actionable problem statement.



Ideate

Generate a wide range of creative solutions without initial judgment.



Prototype

Build low-fidelity versions of ideas to quickly learn and iterate.



Test

Gather feedback from real users to validate and refine the concept.

01. Empathize

Understand the User

Empathy is the foundation of human-centered design. It requires setting aside your own assumptions to gain deep insights into users and their needs.

By conducting interviews and mapping empathy, we uncover not just what people do, but how they think and feel.

Methods for Empathy

Observation & Shadowing: Watch users in their natural environment to see pain points that they might not even notice themselves.

Deep Dive Interviews: Ask "Why?" repeatedly to move past surface-level answers and discover latent needs and motivations.

Immersion: Step into the shoes of the user by experiencing their journey firsthand, feeling the frustrations and successes they face.

02. Define

Framing the Challenge

In this phase, you synthesize your research from the Empathize stage to define the core problem you are trying to solve.

The goal is to create a "Point of View" (POV) that is human-centered, focusing on specific users and their needs rather than business goals.

> *"How Might We encourage healthier eating habits for busy office workers during their lunch break?"*

03. Ideate

Expanding the Possible

Now that the problem is defined, it's time to generate solutions. The key to successful ideation is volume over quality in the beginning.

We use creative triggers and brainstorming rules to bypass self-censorship and encourage wild ideas that might lead to breakthroughs.

Defer judgment.

Encourage wild ideas.

Build on the ideas of others.

Ideation Techniques

Brainstorming

A collaborative, verbal method where the group discusses and builds ideas together out loud. Great for building momentum and energy in the room.

Pro: High energy and immediate feedback.

Brainwriting

Individuals write ideas silently on sticky notes before sharing. This prevents "groupthink" and ensures everyone's voice is heard equally.

Pro: Eliminates bias and captures diverse thoughts.

04. Prototype

Building to Learn

Prototyping is about making ideas tangible. You don't need a high-tech lab; you just need to create something users can interact with.

Low-fidelity prototypes—sketches, paper models, or wireframes—allow for rapid iteration without heavy investment of time or money.

The motto: "Fail fast to learn faster."

05. Test

Validation & Feedback

Testing is where you put your prototype in front of real users. Observe their interactions, listen to their feedback, and watch for friction points.

This isn't about selling your idea; it's about validating if your solution actually solves the problem you defined in Stage 2.

Insights gained here often lead to refining the prototype or even redefining the problem entirely.

The Iterative Cycle

Design thinking is non-linear. Insights from one stage often trigger a return to an earlier one.

Insights

User research reveals a new need.

Redefine

Adjust the POV based on data.

Iterate

Pivot the solution strategy.

Refine

Update prototypes for clarity.

Launch

Implement with user confidence.

Stages in Design Thinking

- Stages:
  1. Empathize
  2. Define
  3. Ideate
  4. Prototype
  5. Test

Stages in Design Thinking

- Stages: Empathize, Define, Ideate, Prototype, Test

Lecture 3

- Human Cantered Software Design

Human Centred Software Design

- Human-Centred Software Design (HCD) shifts the focus of software engineering from technology-first (What can we build?) to people-first (What do users actually need, and how can we make it effortless for them?).
- Instead of forcing users to adapt to how software works, HCD forces the software to adapt to human behavior, cognitive limits, and emotions.

Human Centred Software Design

- **The 4 Core Principles of HCD**
- Focus on People:
  - Always design with a deep empathy for the actual end-users—understanding their environment, daily frustrations, and workflows.
- Fix the Root Problem:
  - Don't just build what users say they want; observe them to discover the true, underlying friction they face.

Human Centered Software Design

- The 4 Core Principles of HCD
- Everything is a System:
  - Recognize that software doesn't exist in a vacuum. A user's experience includes their device, their internet speed, their physical surroundings, and their organizational processes.
- Iterate Continuously:
  - Accept that your first design won't be perfect. Test early, fail fast, and refine continuously based on real human feedback.

Human Centred Software Design

- Human-Centred Software Design (HCSD) is an approach to designing software that
  - Prioritizes the needs
  - Goals,
  - Abilities,
  - Experiences of the people who will use it.
- Instead of starting with technology or features, it begins by
  - Understanding users
  - Designing solutions that are useful,
  - Usable
  - Accessible.

Human Cantered Software Design

- **Key Characteristics**
  - User-focused
  - Empathy-driven
  - Iterative
  - Evidence-based
  - Accessible
  - Collaborative

Human Centred Software Design

- **Principles of Human-Centred Software Design**
  - **Understand users** – Research users' needs, behaviours, and challenges through interviews, surveys, observation, and usability studies.
  - **Involve users throughout the process** – Gather user feedback during planning, design, development, and testing.
  - **Focus on usability** – Ensure the software is easy to learn, efficient to use, and minimizes errors.
  - **Iterative design** – Continuously improve the product based on user feedback and testing.
  - **Accessibility and inclusivity** – Design for people with different abilities, backgrounds, languages, and levels of technical experience.

Human Centred Software Design

Case 1:

- Canteen
  - Personas:
    - Students
    - Faculty/ Staff
    - Suppliers
    - Cashiers
    - Counter Staff
    - Cook
    - Table Cleaner

Human Centred Software Design

Case 1:

- Canteen
- Facilities:
  - Too much crowd
  - Cleanliness

- Food Order System:
  - Queue in the canteen for making payment
  - Waiting time for getting food ready
  - Not knowing if your food is ready

Human Centred Software Design

- **Merits:**
  - Improves user satisfaction.
  - Increases productivity and efficiency
  - Reduces user errors and training requirements
  - Enhances accessibility and inclusiveness
  - Lowers development costs by identifying problems early
  - Increases adoption and customer loyalty.

Human Centered Software Design

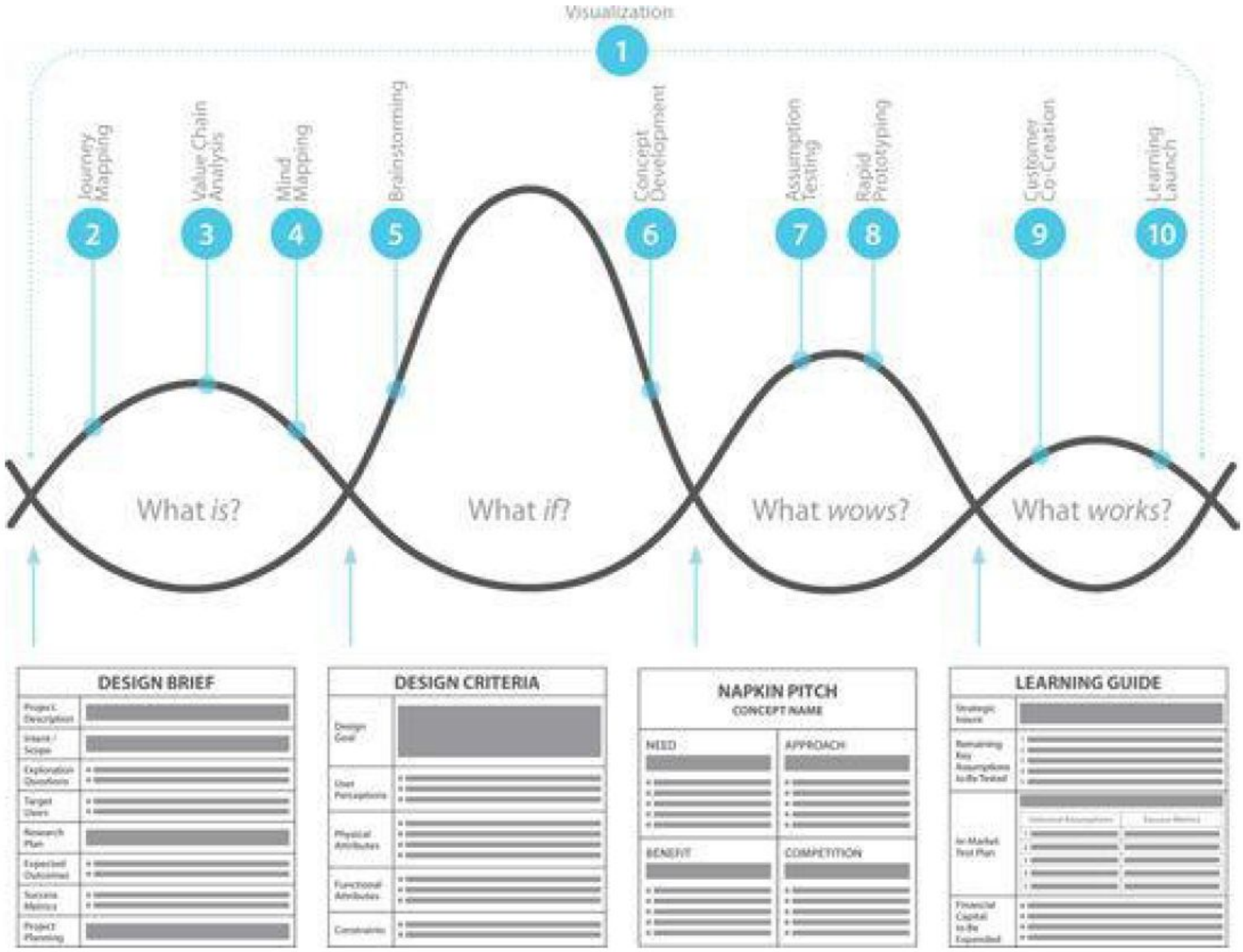
Lecture 6-7

Crystal Ball approach in design

- A strategic foresight exercise where a design team attempts to predict future user behaviours, technological shifts, and market environments 5 to 10 years down the line.
- Like a fortune teller looks into a crystal ball to see the future, designers use trend data, emerging tech signals, and cultural shifts to design products for a world that does not yet fully exist

Comparison

Feature	Crystal Ball	Reactive/ Real Time approach
Data for development	Qualitative macro-trends, foresight signals, futures cones, and expert speculation	Quantitative telemetry, live A/B test data, crash logs, and immediate customer support tickets
Product Lifecycle	Long, slow, and highly strategic research-and-development phases.	Hyper-fast, continuous release cycles (daily or weekly patches and micro-updates).
Risk Profile	**High Risk, High Reward:** You could invent the next iPhone, or you could end up with a Google Glass	**Low Risk, Low Reward:** You rarely fail catastrophically, but you risk making only minor, uninspiring changes.
Example	Automotive Concept Cars	Social Media Platforms


Visualization
1
Journey Mapping
2
Value Chain Analysis
3
Mind Mapping
4
Brainstorming
5
Concept Development
6
Assumption Testing
7
Rapid Prototyping
8
Customer Co-Creation
9
Learning Launch
10
What *is*?
What *if*?
What *wows*?
What *works*?
DESIGN BRIEF
DESIGN CRITERIA
NAPKIN PITCH
CONCEPT NAME
NEED
APPROACH
BENEFIT
COMPETITION
LEARNING GUIDE

PROJECT MANAGEMENT AID	PURPOSE
PMA 1 Design Brief	• Formalizes the growth project • Defines goals, resources, timelines, *etc.* • Serves as a "North Star" throughout the project
PMA 2 Design Criteria	• Sets criteria to evaluate alternative designs (derived from study of user needs and business requirements) • Becomes part of the design brief
PMA 3 Napkin Pitch	• Crystalizes communication of solution concepts (after brainstorming and concept development) • Describes each of the best few (3–5) solutions that meet the design criteria in a template that allows for apples-to-apples comparison
PMA 4 Learning Guide	• Defines an affordable level of resources to invest in learning whether (or not) the top 2–3 concepts are feasible

Approach to Design

10 Tools for Designing

1. Visualization
2. Journey Mapping
3. Value Chain Analysis
4. Mind Mapping
5. Brainstorming
6. Concept Development
7. Assumption Testing
8. Rapid Prototyping
9. Customer co-creation
10. Learning Launch

Approach to Design

1. **Visualization:** using imagery to envision possibilities and bring them to life
2. **Journey Mapping:** assessing the existing experience through the customer's eyes
3. **Value Chain Analysis:** assessing the current value chain that supports the customer's journey
4. **Mind Mapping:** generating insights from exploration activities and using those to create design criteria
5. **Brainstorming:** generating new possibilities and new alternative business models
6. **Concept Development:** assembling innovative elements into a coherent alternative solution that can be explored and evaluated
7. **Assumption Testing:** isolating and testing the key assumptions that will drive the success or failure of a concept
8. **Rapid Prototyping:** expressing a new concept in a tangible form for exploration, testing, and refinement
9. **Customer Co-Creation:** enrolling customers to participate in creating the solution that best meets their needs
10. **Learning Launch:** creating an affordable experiment that lets customers experience the new solution over an extended period of time, to test key assumptions with market data

Approach to Design

1. Visualization

- Integral to the other tools.
- Focus on identifying, organizing and communicating things rather than describing in text and numbers
- Visual imaginary into work process that focuses on bringing idea to life.

Approach to Design

1. Visualization
   - The process of using visual imagery and diagrams to convey ideas and envision possibilities.
   - Shifts team communication away from abstract words and dense text toward concrete visual formats.
   - Using sketches, diagrams, and storyboards makes complex concepts immediately accessible and easier to grasp.
   - Helps bridge communication gaps between multidisciplinary team members who speak different technical languages.
   - Allows teams to bring abstract ideas to life before investing actual resources.

Approach to Design

1. Visualization
   - Stimulates creative thinking and encourages non-linear problem-solving during discussions.
   - Teams can quickly identify missing elements or logical flaws in a concept when laid out visually.
   - Enhances empathy by making customer scenarios and emotional reactions visually apparent.
   - Visual artifacts serve as continuous visual reference points throughout the entire design process.

Approach to Design

2. Journey Mapping
   - An analytical tool used to track and evaluate a customer’s experience over time.
   - Explicitly assesses the current, end-to-end process strictly through the customer's eyes.
   - The map captures every touchpoint, action, and interaction a user has with a product or service.
   - Highlights emotional highs and lows, revealing key pain points and areas of customer frustration.
   - Teams use it to uncover hidden moments of truth where customer loyalty is gained or lost.

Approach to Design

2. Journey Mapping
   - Shifts the internal corporate perspective to an empathetic, customer-centric point of view.
   - Process exposes gaps between what a company thinks it delivers and what users actually experience.
   - Helps prioritize areas that require immediate operational improvement or innovative redesign.
   - Provides context, showing how individual interactions fit into a broader experience.

Lays the groundwork for identifying meaningful opportunities for value creation.

Approach to Design

3. Value Chain Analysis
   - Examines the enterprise operations that support the customer journey.
   - Tries to find out the behind the scenes at the internal capabilities, resources, and operational workflows involved.
   - Focuses on understanding how value is created, delivered, and sustained across partners and suppliers.
   - Identifies which internal processes directly contribute to solving customer pain points.
   - Analysis reveals operational bottlenecks, redundancies, and cost inefficiencies within the organization.

Approach to Design

3. Value Chain Analysis

- Assesses how flexible the current value chain is to support potential new solutions.
- Teams gain clarity on key capabilities required to turn a new design concept into reality.
- Evaluates strategic relationships with external vendors and ecosystem partners.
- The tool ensures that proposed front-end customer experiences are backed by sustainable back-end operations.

Bridges customer-facing desirability with operational and financial feasibility.

Approach to Design

4. Mind Mapping
   - A visual technique used to synthesize insights gathered during research and exploration.
   - Unstructured research data—like user interviews and observations—and organizes it visually.
   - The process displays central themes radiating outward into related sub-ideas and patterns.
   - Allows teams to uncover meaningful relationships, clusters, and underlying customer insights.
   - By connecting disparate data points, mind mapping helps teams make sense of complex qualitative input.

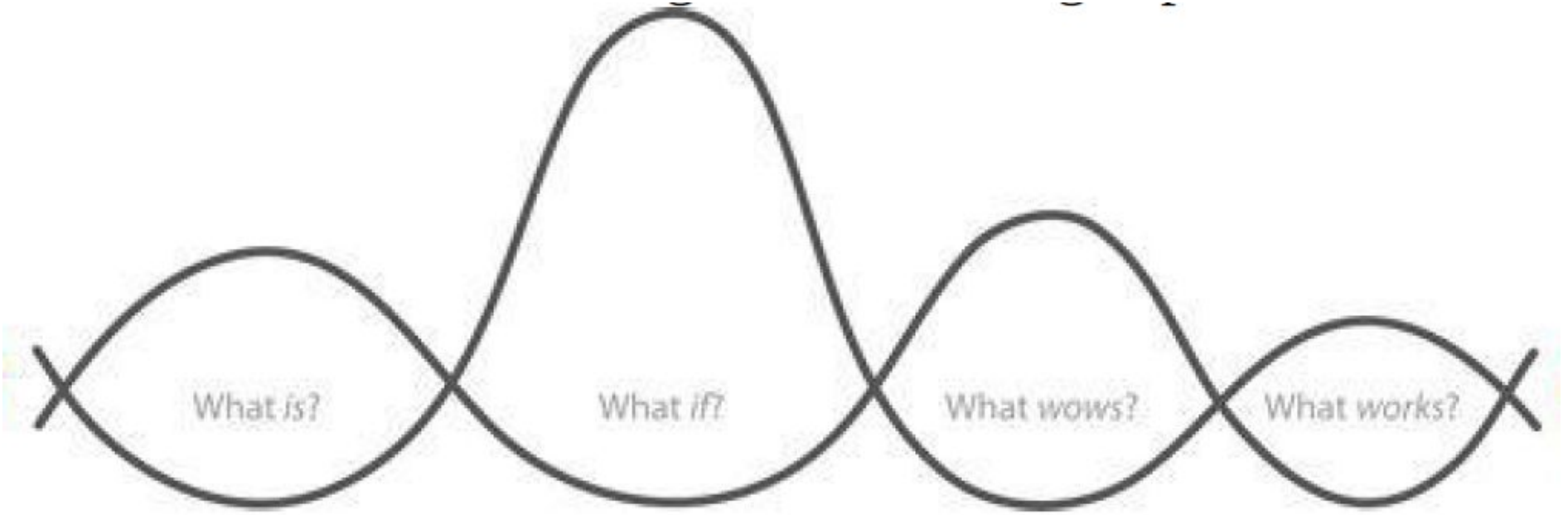
Approach to Design

4. Mind Mapping

- The insights generated from the map are directly used to form actionable design criteria.
- Design criteria act as explicit guidelines defining what a successful future solution must accomplish.
- Prevents teams from rushing to solutions before thoroughly understanding the problem space.
- Keeps the entire research context visible during the transition to ideation.

- **It serves as a bridge between the exploration phase and the generation of creative ideas.**

Design Process



Current Reality **Visualise New future** **Make choices for better solution** **Feasible solution**

Approach to Design

The four stage process:

What ***is?***

What ***if?***

What ***wows?***

What ***works?***

Use *for each of the above questions: broadening* ***(diverging) progressively expanding our field of vision thinking of multiple options*** *and narrowing* ***(conversing) around narrowing down options to most promising***

Reactive / Continuous Deployment approach in design

- Focuses entirely on **the present moment, real-time data, and immediate user behaviour**.
- Find what people face issues, or looking for solution and provide the solution

Design v/s traditional problem solving

Lecture 8-9

- Case studies

Case Studies on Software Design Thinking in software

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Module 1: Introduction to Software Design Thinking

- **SCAMPER**
- A powerful acronym framework used to modify existing products into new innovations:
- **S**ubstitute (What materials or components can we swap?)
- **C**ombine (Can we merge this product with another service?)
- **A**dapt (How can we tweak this for a completely different industry?)
- **M**odify/Magnify (What happens if we make a core feature 10x bigger?)
- **P**ut to another use (How would a child or an elder use this product?)
- **E**liminate (What can we remove to simplify the interface entirely?)
- **R**everse (What if we flip the user journey upside down?)